

KIMBLE COUNTY AP, TX, USA

WMO: 747400

Lat: 30.511N

Lon: 99.766W

Elev: 1749

StdP: 13.79

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.4	25.8	7.0	8.4	39.7	11.1	10.3	40.8	22.1	59.7	19.6	58.0	3.9	290	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.3	100.4	70.3	98.4	70.3	96.4	70.4	75.4	87.6	74.6	86.7	73.8	85.9	9.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	129.0	78.8	71.8	125.3	78.6	70.8	121.0	78.4	40.2	87.1	39.3	86.3	38.6	85.4	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	17.5	16.0	16.2	103.7	3.5	2.7	13.7	105.7	11.6	107.3	9.7	108.8	7.1	110.8
			14.5	77.0	3.3	1.4	12.2	77.9	10.3	78.7	8.4	79.5	6.1	80.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	47.9	52.4	59.4	67.7	75.1	81.3	83.3	83.6	76.8	67.3	56.8	48.9
	DBStd	14.84	8.82	9.97	9.43	7.77	6.55	4.18	3.74	3.64	5.76	8.03	8.76	8.77
	HDD50	427	146	82	29	2	0	0	0	0	0	3	35	130
	HDD65	2032	533	361	221	62	8	0	0	0	4	72	268	503
	CDD50	6547	80	150	320	531	778	939	1033	1041	803	538	239	95
	CDD65	2676	2	9	47	141	321	489	568	576	356	142	23	2
	CDH74	31548	69	184	533	1560	3372	5764	7123	7280	3874	1476	243	70
	CDH80	15774	11	43	153	584	1527	2991	3952	4114	1843	503	43	10
(14) Wind	WSAvg	6.7	6.0	6.8	7.4	8.2	8.1	7.7	6.9	6.0	5.5	5.8	5.8	5.8
(15) Precipitation	PrecAvg	24.1	1.1	1.5	1.8	2.2	2.8	2.8	2.0	2.1	2.6	2.3	1.6	1.2
	PrecMax	35.8	2.9	4.9	4.8	4.0	6.5	7.2	6.3	4.8	5.9	7.0	8.8	5.9
	PrecMin	16.3	0.2	0.0	0.0	0.3	0.6	0.3	0.0	0.1	0.4	0.2	0.0	0.1
	PrecStd	5.5	0.8	1.3	1.3	1.2	1.5	1.8	1.7	1.5	1.5	1.7	2.0	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.2	85.4	88.6	96.1	99.8	102.0	102.4	102.9	100.5	91.9	85.0	80.6
		MCWB	56.5	58.9	61.6	63.5	68.0	71.3	70.3	70.4	68.9	68.0	64.0	57.6
	2%	DB	74.9	79.3	84.3	90.6	95.3	98.5	99.6	100.2	95.6	88.5	80.0	75.0
		MCWB	55.8	57.6	60.9	64.1	69.2	70.7	70.3	70.9	69.9	67.9	62.4	57.8
	5%	DB	70.2	74.9	80.1	86.3	91.7	95.5	97.6	98.2	92.6	85.6	76.5	70.4
		MCWB	55.6	57.2	60.2	64.2	69.1	70.9	70.5	70.9	69.5	66.3	61.0	57.3
	10%	DB	65.6	70.3	75.6	82.6	88.3	92.9	95.4	95.9	89.9	82.4	72.8	66.2
		MCWB	53.5	56.5	59.7	63.0	68.6	71.0	70.8	70.7	69.3	65.7	60.1	55.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	66.0	67.6	71.3	75.4	77.1	76.1	76.4	75.3	74.1	69.6	65.5
		MCDB	70.3	73.1	77.9	83.1	87.7	90.3	89.0	88.9	85.0	83.0	78.8	71.6
	2%	WB	61.3	63.9	66.1	69.6	73.7	75.5	74.9	75.0	74.1	72.3	67.3	63.2
		MCDB	69.0	70.8	75.5	80.4	85.7	87.9	87.5	87.4	84.2	81.0	73.6	69.3
	5%	WB	58.5	61.8	64.6	68.4	72.5	74.4	74.0	74.0	73.2	70.9	65.1	60.0
		MCDB	66.2	69.2	73.5	78.7	84.3	86.8	86.4	86.8	83.5	79.7	71.9	66.8
	10%	WB	54.8	59.2	63.0	67.3	71.4	73.4	73.2	73.2	72.3	69.2	63.0	57.0
		MCDB	63.8	67.7	71.7	77.2	83.0	85.7	85.5	86.3	82.6	78.0	70.2	64.1
(35) Mean Daily Temperature Range	5% DB	MDBR	27.2	26.2	25.6	25.8	22.6	21.9	22.3	23.3	23.1	24.8	25.7	26.5
		MCDBR	36.0	33.6	30.8	30.5	27.6	25.8	25.5	26.5	27.0	28.3	30.2	33.5
	5% WB	MCWBR	20.2	18.0	14.8	13.1	9.2	6.7	6.2	6.2	8.2	12.0	15.8	19.0
		MCWBR	25.9	23.9	22.8	21.9	21.4	20.5	21.5	22.3	20.2	19.9	21.2	24.4
(40) Clear-Sky Solar Irradiance	taub		0.303	0.308	0.332	0.372	0.400	0.403	0.415	0.403	0.385	0.351	0.322	0.306
		taud	2.534	2.524	2.468	2.375	2.354	2.395	2.408	2.428	2.414	2.498	2.511	2.530
	Ebn at Noon		297	305	302	292	282	280	276	279	281	285	287	290
		Edh at Noon	27	30	34	39	40	38	37	36	36	30	28	26
(44) All-Sky Solar Radiation	RadAvg		1034	1253	1571	1870	1948	2200	2158	2021	1668	1376	1077	930
	RadStd		110	173	137	158	202	171	182	112	148	196	141	123

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.92	N/A	N/A	N/A	+0.88	+1.21	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page